The shortest relations of planar triangle reflection groups

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August 10, 2026

Abstract

Let τ be a Euclidean triangle and G_τ the group generated by the three reflections in its sides. We determine the shortest relations satisfied by these generators for every τ: there are exactly 33 of them, each equating two reduced words of length 11, and for all τ outside a proper Zariski-closed set of shapes they are the only coincidences in the ball of radius 11. The associated growth sequence begins $1, 3, 6, 12, 24, 48, 96, 192, 384, 768, 1536, 3039, 6012$. We complement this with the word metric of the translation subgroup: its elements are signed configurations of lamps on a hexagonal site lattice, and we identify them with the finitely supported integer flows on the honeycomb Cayley graph of the point group $\mathbb{Z}^2 \rtimes C_2$. For all shapes outside countably many proper subvarieties, the word length of a configuration equals the ℓ^1-norm of its flow plus twice the least number of edges connecting the support of the flow to the base vertex. That length formula is not new (it is the formula of Droms, Lewin and Servatius and of Myasnikov, Roman'kov, Ushakov and Vershik for the groups F/N', the images of the Magnus embedding) and what is established here is the identification which places G_τ in their setting, together with the small difference caused by our generators being involutions. The cancellation and single-lamp laws follow, and the formula reproduces the exact census of translations through length 18, element by element. The shape space is stratified: on the lines where one angle equals π/m the relation onset moves to depth $6, 8, 9, 10, 10$ for $m = 4, 6, 3, 2, 8$ and is the universal one for the remaining $m \leq 12$, so that the crystallographic orders are not the distinguished ones. We prove that each divisor d of m with $2 \leq d \leq m/2$ contributes $d^2 - 1$

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relations at depth $m+m/d$, which for even m gives $(m/2)^2-1$ relations at depth $m+2$.

Keywords. triangle reflection group; growth series; word metric; Coxeter group; lamplighter group; crystallographic restriction.

2020 Mathematics Subject Classification. Primary 20F55, 20F65; Secondary 20F05, 51F15, 05A15.

1 Introduction

Let $\tau \subset \mathbb{R}^2$ be a nondegenerate Euclidean triangle and let r_0, r_1, r_2 be the reflections in the lines containing its three sides. They generate a group $G_\tau \leq \mathrm{Isom}(\mathbb{R}^2)$, the *triangle reflection group* of τ. When the angles of τ are of the form π/n_i the triangle tiles the plane and G_τ is a Euclidean reflection group with the classical finite presentation. For all other shapes the group is not crystallographic, and Isola and Piergallini [7] showed that for a *generic* triangle, meaning one whose edge lengths are algebraically independent, the translation subgroup T_τ is free abelian of infinite rank, the orientation preserving subgroup is free metabelian of rank two, and G_τ admits no finite presentation. Their work determines the relations of G_τ as a group. It does not address the word metric.

The present paper takes up the metric question. Write $F = C_2 * C_2 * C_2 = \langle x_0, x_1, x_2 \mid x_i^2 \rangle$ and let $\rho_\tau \colon F \to G_\tau$ send x_i to r_i. Elements of F are reduced words, and the sphere of radius d in F has $3 \cdot 2^{d-1}$ elements for $d \geq 1$. The question is where, and by how much, the image sphere is smaller.

Theorem 1.1 (Census)**.** *There is an explicit list of* 33 *pairs* (w_i, w_i') *of reduced words of length* 11*, pairwise distinct in* F*, such that*

(i) $\rho_\tau(w_i) = \rho_\tau(w_i')$ *for every nondegenerate Euclidean triangle* τ *and every* i*;*

(ii) for τ *outside a proper Zariski-closed subset of the shape space, these are the only coincidences of* ρ_τ *on the ball of radius* 11*, and on the ball of radius* 12 *the only further coincidences are the* 132 *described in Remark 2.1.*

Consequently the growth sequence $u_d(\tau) = \#\{g \in G_\tau : \ \ell(g) = d\}$ *of the word metric of a generic triangle begins*

$$1,\ 3,\ 6,\ 12,\ 24,\ 48,\ 96,\ 192,\ 384,\ 768,\ 1536,\ 3039,\ 6012,$$

its first deviation from $3 \cdot 2^{d-1}$ *occurring at* $d = 11$ *with deficit* 33*, and the shortest elements of* $\ker \rho_\tau$ *common to all triangles have length exactly* 22*.*

The second half of the paper concerns T_τ. By [7, Thm. 4.1] it is free abelian on the conjugacy class of $t_1 = (r_0 r_1 r_2)^2$, so its elements are finitely supported integer vectors on a lattice of *sites*; we call a nonzero entry a *lamp*. Conjugation by the generators acts on sites through three affine involutions whose pairwise products are translations by the six vectors of a hexagonal lattice (§3). In §5 we identify T_τ, for generic shapes, with the group of finitely supported integer flows on the Cayley graph X of the point group $\mathbb{Z}^2 \rtimes C_2$, a cubic graph whose hexagonal faces correspond to the sites, and determine the metric completely:

Theorem 1.2 (see Theorem 5.7; the length formula is [9, Thm. 2.10] in the present setting)**.** *For flow-generic* τ *(Definition 5.4) and every reduced configuration* c*,*

$$\ell(c) \;=\; \sum_{\{u,v\}} |c_u - c_v| \;+\; 2\,\mathrm{st}(c),$$

the sum over unordered hex-adjacent pairs of sites with c *extended by zero, where* $\mathrm{st}(c)$ *is the least number of edges of* X *whose addition connects the support of the flow of* c *together with the base vertex.*

The first summand is the ℓ^1-norm of the flow of c; the second is a Steiner term. The cancellation rule for lamps, the single-lamp costs of §4, and the exact census of translations by word length all follow (§5).

The formula itself is known, and in greater generality: for a free group F and any normal $N \lhd F$, the word length in F/N' of an element is the ℓ^1-norm of its flow on the Cayley graph of F/N plus twice the size of a minimal connecting forest [9, Thm. 2.10], attributed there and in [11, Lem. 3.2] to [6]. What is proved here is the identification that brings G_τ into that setting (Theorem 5.6, which says that for flow-generic τ the translation subgroup is exactly the cycle space of X and G_τ is the corresponding extension) and the bookkeeping needed because our generators are involutions, so that the relevant graph is the honeycomb X and not the Cayley graph of F_3/N. Both points are taken up in §5.

Section 6 records what happens on the non-generic strata. If one angle is exactly π/m and the shape is otherwise generic, then the reference group is the Coxeter group $\langle x_i \mid x_i^2, (x_0 x_1)^m \rangle$, whose growth series we compute in closed form, and the first deficit occurs at a depth $d^*(m)$ which, for $m \le 12$, is smaller than the universal 11 exactly when $m \in \{2, 3, 4, 6, 8\}$. The crystallographic orders are therefore not the ones admitting early relations, correcting the expectation that the crystallographic restriction governs the stratification: what governs it is the order of the rotations that the stratum creates, through the divisors of m.

A companion paper [5] studies the same family from a different side: it works at one fixed shape, the universal one, and analyses the growth sequence A396406 and the lamplighter structure of its translation subgroup. The overlap with the present paper is the ambient setting and the translation metric of §5, which both papers use; nothing proved here is quoted from there, and the census, the strata and the rotation relations below are independent of it.

2 The census theorem

Every triangle is similar to one with vertices $(0,0)$, $(1,0)$, (p,q) with $q \neq 0$, and similarity conjugates ρ_τ, so all coincidence data depends only on (p,q). Over $\mathbb{Q}(p,q)$ the three reflections have matrix entries that are rational functions whose denominators are the squared side lengths, and these do not vanish for nondegenerate τ.

Proof of Theorem 1.1. For (i): for each of the 33 pairs the identity $\rho(w_i) = \rho(w_i')$ is verified by exact arithmetic in $\mathbb{Q}(p,q)$. An identity of rational functions holds at every nondegenerate specialization, acute or obtuse.

For (ii): at the rational witness τ_0 with third vertex $(1/3, 1/2)$, exact arithmetic over $\mathbb{Q}$ shows that ρ_{τ_0} is injective on the ball of radius 10 and that its only coincidences on the ball of radius 11 are the 33 listed pairs, each consisting of two words of length exactly 11; hence the layer counts are $3{\cdot}2^{d-1}$ for $1 \leq d \leq 10$ and $3072 - 33 = 3039$ at $d = 11$. Any coincidence $\rho_\tau(w) = \rho_\tau(w')$ with $w \neq w'$ of length at most 11 is the vanishing of a rational function of (p,q); by the witness it is not identically zero unless (w,w') is one of the 33 pairs, so each of the finitely many remaining conditions defines a proper closed subset. Off their union the coincidences are exactly the 33 pairs.

Finally, no cancellation occurs in $w_i(w_i')^{-1}$. If w_i and w_i' shared a common suffix, say $w_i = us$ and $w_i' = u's$, then $\rho_\tau(u) = \rho_\tau(u')$ with $u \neq u'$ of length at most 10, contradicting injectivity on that ball; a common prefix is excluded in the same way. Hence each relator has length exactly 22. No shorter element of $\ker \rho_\tau$ is common to all shapes: if z lay in $\ker \rho_\tau$ for every τ and had length $L \leq 21$, write $z = AB^{-1}$ with $|A| \leq 11$ and $|B| \leq 10$, so that $\rho_\tau(A) = \rho_\tau(B)$ with $A \neq B$ at every shape. If $|A| \leq 10$ this contradicts injectivity on the ball of radius 10; if $|A| = 11$ it is a coincidence between a word of length 11 and one of length at most 10, and by (ii) every coincidence has both words of length exactly 11.

For the radius 12 statement, enumeration of all reduced words of length at most 12 at the witness τ_0, and independently at the witness with third vertex $(2/7, 3/5)$, returns the same 132 coincidences beyond those of the ball of radius 11 and no others; each of the 132 is an identity over $\mathbb{Q}(p, q)$, verified after clearing denominators as an identity of polynomials in $\mathbb{Z}[p, q]$. The argument of the previous paragraph applies verbatim to the finitely many remaining conditions on the ball of radius 12. □

Remark 2.1. The 132 coincidences at length 12 carry no new relations: 66 of them are the one letter extensions of the 33 pairs, 33 of the form $w_i x = w_i' x$ and 33 of the form $xw_i = xw_i'$, exactly one letter x on each side keeping both words reduced; and 66 are the splittings of the 33 relators into a word of length 12 and a word of length 10, two per relator. In particular 66 reduced words of length 12 represent elements at distance 10, so the image $\rho_\tau(\{|w| = 12\})$ has $6144 - 66 = 6078$ elements whereas the sphere of the word metric has $6078 - 66 = 6012$; it is the latter that the theorem records, and that the breadth-first computations return. The count, its $66 + 66$ split and the growth sequence through depth 12 are regenerated by `code/zeta_probe/tools/paper4_ball12` at all three witnesses and at two primes.

Remark 2.2. The witnesses $(-1/3, 1/2)$, obtuse, and $(2/7, 3/5)$ produce the same 33 classes. By [7, Thm. 4.2] each of the 33 relators lies in the normal closure of the commutators of t_1 with its conjugates, so the theorem is consistent with, and refines metrically, the presentation given there. Their Section 6 tabulates cyclically reduced stable words up to length 24 modulo symmetries; that census is of the relator language, not of the geodesic ball.

3 Sites, lamps and the hexagonal lattice

Normalize the incircle of τ to be the unit circle at the origin, so that $(x)r_i = (x)s_i + 2v_i$ with s_i the linear part and v_i the unit vector to the tangency point of the ith side [7, §3]. Set $t_{n,m} = (\!(t_1)\!)(r_0 r_1)^n (r_0 r_2)^m$; by [7, Thm. 4.1] these form a basis of T_τ for generic τ. Conjugation by the generators acts on the index by

$$\sigma_0(n, m) = (-n-1, -m+1), \quad \sigma_1(n, m) = (-n-2, -m+1), \quad \sigma_2(n, m) = (-n-1, -m),$$

three affine involutions whose pairwise products are the translations by

$$a=(1,0), \qquad b=(0,1), \qquad c=(1,-1), \qquad a=b+c.$$

We call $\{\pm a, \pm b, \pm c\}$ the *hex generators* and say two sites are *hex-adjacent* if their difference is one of them; the sites $(0,0),(-1,0),(-1,1)$ are the *base sites.*

Proposition 3.1 (Reduction). *Let τ be flow-generic in the sense of Definition 5.4. Then distinct reduced integer vectors on sites represent distinct elements of T_τ. In particular no site carries both signs in a reduced representative, and a configuration containing a lamp together with its inverse at one site represents the identity.*

Proof. Theorem 5.6 gives that T_τ is free abelian on the lamps $t_{n,m}$ and that $(n,m) \mapsto t_{n,m}$ is injective, which is exactly the assertion. We state the proposition here because it is what the notation of this section presumes, but its proof belongs to §5 and does not depend on anything between. □

Remark 3.2. The hypothesis is not decoration. Without it the conclusion is false, and the strata of §6 are where it fails: on the line $\alpha = \pi/m$ one has $t_{n,j} = t_{n+m,j}$, so two distinct reduced vectors, the single lamps at $(0,j)$ and at (m,j), have the same value, and by Theorem 6.6(i) the m lamps of a level even sum to zero.

The statement itself is not new. [7, Thm. 4.1] gives freeness on the conjugacy class of t_1 for a triangle typical in their sense, the angles being $\mathbb{Q}$-linearly independent, and their equation (17) indexes that class by the same double conjugation used here, the index being uniquely determined because τ is typical; so both the freeness and the indexing are theirs, under their hypothesis. What Theorem 5.6 adds is that flow-genericity suffices, a condition that algebraic independence of the coordinates implies and that is not claimed to be equivalent to it (Proposition 5.5).

4 Single lamps

The length of a single lamp is governed by one quantity, the distance in X from the base vertex to the face of the site. For a site (n,j) write

$$k(n,j) \;:=\; d_X\big(e, f_{n,j}\big),$$

the graph distance in X from the base vertex e to the vertex set of the face $f_{n,j}$ of (n,j), in the notation of §3. The exact length formula $\ell(t^{\pm 1}_{n,j}) =$

$6+2k(n,j)$ is proved for flow-generic shapes in Corollary 5.8, once the metric of §5 is available, and the closed form of k itself is Lemma 6.7. The present section fixes the notation only.

5 The translation metric

Write $F = C_2 * C_2 * C_2$ as before and let

$$\widehat{P} \;=\; \big\langle\, x_0, x_1, x_2 \;\big|\; x_i^2,\; [x_0x_1,\, x_0x_2]\,\big\rangle.$$

The rotations $\rho_1 = x_0x_1$ and $\rho_2 = x_0x_2$ commute in $\widehat{P}$ and are inverted by conjugation with x_0, so every element has the normal form $\rho_1^{m_1}\rho_2^{m_2}x_0^{\varepsilon}$ and $\widehat{P} \cong \mathbb{Z}^2 \rtimes C_2$. Let X be the left Cayley graph of $\widehat{P}$ on x_0, x_1, x_2, with edges $\{g, x_ig\}$ and base vertex e: a connected cubic simple graph, bipartite for the orientation character $\epsilon(x_i) = -1$. For every nondegenerate τ the linear parts satisfy $s_i^2 = 1$ and $[s_0s_1, s_0s_2] = 1$, plane rotations commuting, so $x_i \mapsto s_i$ defines a homomorphism $M_\tau\colon \widehat{P} \to \mathrm{O}(2)$ at every shape.

5.1 Walks, flows, deposits

For a word $w = x_{i_1}\cdots x_{i_\ell}$ let $\sigma_j \in \widehat{P}$ be the image of the suffix $x_{i_{j+1}}\cdots x_{i_\ell}$, so that $\sigma_\ell = e$ and $\sigma_{j-1} = x_{i_j}\sigma_j$. The *suffix walk* $\sigma_\ell, \ldots, \sigma_0$ traverses at step j the edge $\{\sigma_j, x_{i_j}\sigma_j\}$ of X. Orient each edge of X from its $\epsilon = +1$ endpoint and let the *flow* $\phi(w) \in \mathbb{Z}^{E(X)}$ be the net signed traversal count of the suffix walk. When w maps to 1 in $\widehat{P}$ the walk is closed and $\phi(w)$ is a *cycle*: at each vertex the signed values of the three incident edges sum to zero. Flow is additive under concatenation of words with trivial image.

In the incircle expansion of §3 the translation part of $\rho_\tau(w)$ is

$$t(w) \;=\; \sum_{j=1}^{\ell} 2v_{i_j}\, s_{i_{j+1}}\cdots s_{i_\ell} \;=\; \sum_{j=1}^{\ell} 2v_{i_j}\, M_\tau(\sigma_j).$$

Step j deposits $2v_{i_j}M_\tau(\sigma_j)$, and the deposits of the two directions of one edge are opposite: $v_is_i = -v_i$ gives $2v_iM_\tau(x_i\sigma) = -2v_iM_\tau(\sigma)$. Writing $d_\tau(E)$ for the deposit of the canonical direction of the edge E,

$$t(w) \;=\; V_\tau\big(\phi(w)\big), \qquad V_\tau(\phi) \;=\; \sum_{E\in E(X)} \phi_E\, d_\tau(E), \tag{1}$$

for every shape and every word with trivial image.

5.2 Faces

Let $w_6 = (x_0x_1x_2)^2$, the word of t_1.

Lemma 5.1. *Modulo the relations x_i^2 one has $w_6 = [\rho_1, \rho_2^{-1}]$; consequently $\ker(F \to \widehat{P})$ is the normal closure of w_6, and the flows of its elements form the $\mathbb{Z}$-span of the flows of the conjugates $uw_6^{\pm 1}u^{-1}$.*

Proof. $x_0x_1x_2x_0x_1x_2 = (x_0x_1)(x_2x_0)(x_1x_0)(x_0x_2) = \rho_1\rho_2^{-1}\rho_1^{-1}\rho_2$. Hence imposing $w_6 = 1$ on the free product imposes exactly the commutation of ρ_1 and ρ_2, which presents $\widehat{P}$. A product of conjugates of $w_6^{\pm 1}$ is a concatenation of words with trivial image, on which flow is additive. $\square$

Lemma 5.2 (Faces)**.** *The support of $\phi(w_6)$ is a hexagonal cycle through e, and $\phi(uw_6^{\pm 1}u^{-1}) = \pm\,\phi(w_6) \cdot g^{-1}$, where g is the image of u and $\widehat{P}$ acts on X by right translation. The translated hexagons, the* faces *of X, are indexed by the sites of §3 through the dictionary $t_{n,m}$; write ∂f_s for the oriented flow of the face at site s. Then:*

(i) every vertex of X lies on exactly three faces, the faces at e being those of the three base sites, and every edge lies on exactly two;

(ii) two faces share an edge exactly when their sites are hex-adjacent, and then they share one edge, with opposite orientations;

(iii) the ∂f_s are linearly independent, and the flows of the elements of $\ker(F \to \widehat{P})$ are precisely the finite sums $\sum_s c_s\,\partial f_s$ with $c_s \in \mathbb{Z}$.

Proof. In the suffix walk of $uw_6^{\pm 1}u^{-1}$ the suffixes of the u^{-1} segment retrace those of the u segment in the opposite direction, so their traversals cancel, while the suffixes of the middle segment are those of $w_6^{\pm 1}$ multiplied on the right by g^{-1}; this is the displayed formula. The suffix walk of w_6 visits six distinct elements of $\widehat{P}$ and closes, giving a hexagon through e. Conjugation permutes the lamps through the involutions σ_i of §3, which indexes the orbit of hexagons by the sites. Right translation is a graph automorphism acting transitively on vertices, so (i) follows from its validity at e, where exact computation shows the three faces through e are those of the base sites $(0,0), (-1,0), (-1,1)$ and each of the three edges at e lies on two of them. For (ii), faces sharing an edge have intersecting vertex sets, which confines the difference of their sites to a finite set; exact computation over this set shows a common edge occurs precisely at the six hex differences, is then unique, and carries opposite orientations in the two faces. For (iii), spanning is Lemma 5.1. Independence: order sites lexicographically and let

s_0 be maximal among the sites s with $c_s \neq 0$ in $\sum_s c_s \partial f_s$; the edge shared by f_{s_0} and f_{s_0+a} receives $\pm c_{s_0} \neq 0$, since $c_{s_0+a} = 0$. $\square$

5.3 Separation and genericity

Lemma 5.3 (Separation). *For every nonzero finitely supported cycle ϕ on X, the set of shapes (p, q) with $V_\tau(\phi) = 0$ is a proper Zariski-closed subset of the shape plane.*

Proof. The entries of $V_\tau(\phi)$ are rational functions of (p, q), so it suffices to show that $V_\tau(\phi)$ does not vanish identically. Every edge E has one endpoint with $\epsilon = +1$; write its normal form $\rho_1^{m_1}\rho_2^{m_2}$ and let i be the letter of E. Let β_1, β_2 be the rotation angles of $M_\tau(\rho_1), M_\tau(\rho_2)$ and θ_i the direction of the ith side, so that $\theta_1 \equiv \theta_0 - \beta_1/2$ and $\theta_2 \equiv \theta_0 - \beta_2/2$ modulo π, up to one global choice of sign of the β_k. The deposit $d_\tau(E)$ is a nonzero vector in the direction $\theta_i + \frac{\pi}{2} + \eta\,(m_1\beta_1 + m_2\beta_2)$ modulo π, with $\eta = \pm 1$ one global sign. The *frequency* of E, the coefficient vector of (β_1, β_2) in this direction, lies in $(\frac{1}{2}\mathbb{Z})^2$; doubled, the frequencies of edges with letter $0, 1, 2$ are congruent to $(0, 0), (1, 0), (0, 1)$ modulo 2 respectively. With the two global signs fixed, the letter of E is determined by the parity of its doubled frequency and the exponents (m_1, m_2) by its integer part, so distinct edges have distinct frequencies. If $V_\tau(\phi)$ vanished identically then, the map from shapes to (β_1, β_2) being open near a suitable shape (indeed $(p, q) \mapsto (\beta_1, \beta_2)$ is a diffeomorphism onto $\{\beta_1, \beta_2 > 0,\ \beta_1 + \beta_2 < \pi\}$, since two angles determine the triangle up to similarity), a finite sum $\sum_E \pm\phi_E\,|d_\tau(E)|\,e^{i(c_E + f_E\cdot\beta)}$ with distinct frequencies f_E would vanish on a nonempty open set of β; substituting $z_k = e^{i\beta_k/2}$ makes it a Laurent polynomial vanishing on an open subset of the torus, so all coefficients vanish, and $|d_\tau(E)| \neq 0$ forces $\phi = 0$.

Two facts are used here that the display does not show. First, $|d_\tau(E)| = 2$ for every edge, because §3 normalises the incircle to the unit circle and each v_i is a unit vector; the coefficients of the Laurent polynomial are therefore independent of β, without which the last step fails. Second, the sign $\pm$ must be locally constant in the shape, which it is: the outward normal varies continuously on a connected component of shape space, and vanishing on a nonempty open subset of one component is all the argument needs. Working with directions modulo π, as we do above, discards exactly this information, so we record it here. $\square$

The word *generic* is used in this paper in three distinct senses, and we fix them now. A shape is *generic* when its coordinates are algebraically independent over $\mathbb{Q}$, which is the sense of §2 and of [7]; it is *flow-generic* in the sense of Definition 5.4 just below, which is the hypothesis of the metric theorem; and it is *stratum-generic* in the sense of Definition 6.5, which is the corresponding notion inside a fixed stratum. Algebraic independence implies the other two, by Proposition 5.5 and its stratum analogue, but the converses are not claimed.

Definition 5.4. The shape τ is *flow-generic* if M_τ is injective and $V_\tau(\phi) \neq 0$ for every nonzero finitely supported cycle ϕ on X.

Proposition 5.5. *Flow-genericity fails only on a countable union of proper Zariski-closed subsets of the shape plane. In particular every τ whose coordinates p, q are algebraically independent over $\mathbb{Q}$ is flow-generic.*

Proof. Injectivity of M_τ reduces to injectivity on the rotation subgroup, reflections never being trivial, and there it is the condition $m_1\beta_1 + m_2\beta_2 \notin 2\pi\mathbb{Z}$ for $(m_1, m_2) \neq (0, 0)$. Each failure is a proper condition: the two base angles determine the triangle up to similarity, so $(p, q) \mapsto (\beta_1, \beta_2)$ is a diffeomorphism onto the open set $\{\beta_1, \beta_2 > 0,\ \beta_1 + \beta_2 < \pi\}$, and a nonzero linear functional $m_1\beta_1 + m_2\beta_2$ is nonconstant there, so each level set $\{m_1\beta_1 + m_2\beta_2 = 2\pi k\}$ is a proper closed subset. There are countably many such conditions. With Lemma 5.3 this exhausts the conditions, each defined over $\mathbb{Q}$; a point with algebraically independent coordinates lies on no proper $\mathbb{Q}$-defined Zariski-closed subset. □

Theorem 5.6 (Structure)**.** *Let τ be flow-generic. Every word representing a translation has closed suffix walk; all words representing the same translation have the same flow; and $c \mapsto \phi(c) := \sum_s c_s\, \partial f_s$ identifies the finitely supported integer vectors on sites both with the cycles of Lemma 5.2(iii) and with T_τ. In particular T_τ is free abelian on the lamps, recovering [7, Thm. 4.1] for flow-generic shapes, and*

$$\|\phi(c)\|_1 \;=\; \sum_{\{u,v\}} |c_u - c_v|, \tag{2}$$

the sum over unordered hex-adjacent pairs of sites with c extended by zero.

Proof. A word w with $\rho_\tau(w) \in T_\tau$ has trivial linear part, hence trivial image in $\widehat{P}$ by injectivity of M_τ, and $\phi(w)$ lies in the span of the faces by Lemma 5.2(iii). If w, w' represent the same translation then $V_\tau(\phi(w) - \phi(w')) = 0$ by (1), so $\phi(w) = \phi(w')$ by flow-genericity. Taking products of conjugated lamp words matches the flow of the configuration c with $\sum_s c_s \partial f_s$, injectivity of $c \mapsto \phi(c)$ is Lemma 5.2(iii), and surjectivity onto

T_τ follows since every translation is represented by some word. For (2): every edge of the face of u is shared with exactly one neighbouring face v by Lemma 5.2(i), and carries the flow $\pm(c_u - c_v)$ by (ii). $\square$

5.4 The metric theorem

Theorem 5.6 is what does the work here: it places T_τ, for flow-generic τ, in a setting where the word metric is already known. Let F be free on $x_1, \ldots, x_r$, let $N \lhd F$ be *any* normal subgroup, and let Γ be the Cayley graph of F/N on the images of the x_i. A word $w \in F$ traces a path p_w in Γ from the base vertex and so induces a flow π_w, and the word length of w in F/N' is

$$\ell(w) \;=\; \sum_{e \in \operatorname{supp} \pi_w} |\pi_w(e)| \;+\; 2\,|E(Q)|,$$

with Q a set of edges of least cardinality making $\operatorname{supp} \pi_w \cup \{1, w^\mu\}$ connected: a *minimal forest* for w. This is [6, Thm. 2], reproved with the minimal-forest formulation as [9, Thm. 2.10, with eqs. (6),(7)] and restated as [11, Lem. 3.2]; in all three, N is an arbitrary normal subgroup, so F/N is an arbitrary finitely generated group and no commutativity is assumed. (The separate polynomial-time assertion of [6, Thm. 1] is shown to be fallacious in [9, §4], which proves instead that the problem is NP-complete; the length formula is untouched by this.) The group F/N' is the image of the Magnus embedding $F/N' \hookrightarrow \mathbb{Z}^r \wr F/N$ [8], whose metric behaviour is the subject of [11, 12]. The identification of N/N' with the first homology of Γ is already in [6]; presenting the group itself as an extension of F/N by that homology with a distinguished cocycle (which for us is Theorem 5.6) goes back to [13]. For wreath products proper the connectivity term is instead of travelling-salesman type, the lamps sitting on vertices rather than on edges [10].

Our generators are involutions in $\widehat{P}$, and this places us just outside the quoted setting. In Γ the letters x and x^{-1} traverse two distinct edges between the same pair of vertices whenever $\bar{x}^2 = 1$, so Γ carries a bigon wherever X carries an edge, and $H_1(\Gamma; \mathbb{Z})$ acquires one class per bigon: the class of x^2, which is trivial in G_τ but not in F_3/N'. Thus G_τ is the quotient of F_3/N' by the $\mathbb{Z}[\widehat{P}]$-submodule generated by those classes, and X is the honeycomb rather than Γ. The adaptation is routine and we claim nothing new for it; we give the direct proof because it is three lines in each direction, is uniform in the ambient graph, and is what the Lean formalization recorded in the Declarations follows.

Theorem 5.7 (Metric)**.** *Let c be a nonzero reduced configuration with flow $\phi = \phi(c)$, and let* $\operatorname{st}(c)$ *be the least cardinality of a set M of edges of X*

outside the support of ϕ *such that the subgraph of* X *consisting of the edges* $\operatorname{supp}\phi \cup M$*, their endpoints, and the vertex* e *is connected. Then*

$$\ell(c) \;=\; \|\phi(c)\|_1 \;+\; 2\,\mathrm{st}(c)$$

for flow-generic τ*, and* $\ell(c) \le \|\phi(c)\|_1 + 2\,\mathrm{st}(c)$ *for every* τ*. In particular* $\ell(c) \equiv \|\phi(c)\|_1 \pmod 2$*, and every translation has even length.*

Proof. Lower bound. Let w be any word representing c and let n_E count all traversals of the edge E by its suffix walk, so that $\ell(w) = \sum_E n_E$, and, the two directions of an edge contributing opposite signs, $n_E \ge |\phi_E|$ and $n_E \equiv \phi_E \pmod 2$, where $\phi = \phi(w) = \phi(c)$ by Theorem 5.6. The traversed edges form a connected subgraph containing e, consecutive steps sharing a vertex and the walk beginning at e. Let M be the set of traversed edges with $\phi_E = 0$; each has $n_E \ge 2$ by parity, the set $\operatorname{supp}\phi \cup M \cup \{e\}$ is connected, so $|M| \ge \mathrm{st}(c)$, and

$$\ell(w) \;=\; \sum_{\operatorname{supp}\phi} n_E + \sum_M n_E \;\ge\; \|\phi\|_1 + 2|M| \;\ge\; \|\phi\|_1 + 2\,\mathrm{st}(c).$$

The congruence follows by summing $n_E \equiv \phi_E \pmod 2$, and evenness from (2), each site having six incident lattice edges.

Upper bound. Let M attain $\mathrm{st}(c)$. Form the directed multigraph with $|\phi_E|$ parallel copies of each support edge, directed by the sign of ϕ_E, and one copy of each direction of each edge of M. Every vertex is balanced: the cycle condition at a vertex is the balance of the signed values of its incident edges, and the doubled edges of M are balanced. The multigraph is connected, contains e, and e is incident to an edge, so it carries an Eulerian circuit through e; reading the letters of its edges in reverse order yields a word whose suffix walk is the circuit, of length $\|\phi\|_1 + 2|M|$, with flow ϕ and trivial image in $\widehat{P}$. Its linear part is trivial and its translation part is $V_\tau(\phi(c))$ by (1), so it represents c, at every shape. □

Corollary 5.8 (Single lamps)**.** *For flow-generic* τ*,* $\ell(t_{n,m}^{\pm 1}) = 6+2\,d_X(e, f_{n,m})$*, the graph distance in* X *from the base vertex to the vertex set of the face of* (n, m)*, which is the quantity* $k(n, m)$ *of §4.*

Proof. Here $\|\phi\|_1 = 6$. A geodesic from e to the nearest vertex of the face uses no face edge and connects, so $\mathrm{st} \le d_X(e, f_{n,m})$; conversely the connected graph $\operatorname{supp}\phi \cup M \cup \{e\}$ contains a path from e to the face whose edges, up to its first face vertex, all lie in M, so $\mathrm{st} \ge d_X(e, f_{n,m})$. The agreement with k is exact computation. □

Corollary 5.9 (Cancellation). $\|\phi(c)\|_1 = 6\sum_s |c_s| - 2\sum_{\{u,v\}} \min(|c_u|, |c_v|)$, *the second sum over hex-adjacent pairs of sites carrying equal signs. For configurations without repeated lamps this is* $6L - 2h$, *with* L *the number of lamps and* h *the number of same-sign hex-adjacent site pairs.*

Proof. On the edge between adjacent sites u, v one has $|c_u - c_v| = |c_u| + |c_v|$ when the signs differ or one side vanishes, and $|c_u| + |c_v| - 2\min(|c_u|, |c_v|)$ when they agree; sum over the six edges of each site and halve the double count. □

Corollary 5.10 (Connected stratum). $\mathrm{st}(c) = 0$ *if and only if the support of* $\phi(c)$ *together with* e *is connected, and then* $\ell(c) = \sum_{\{u,v\}} |c_u - c_v|$.

Two families illustrate the connectivity hypothesis. For the configuration with one lamp of a common sign at each of the three base sites, the three edges at e are the edges shared by the three base faces and their flows cancel, so e is isolated from the support: $\mathrm{st} = 1$ and $\ell = 12 + 2 = 14$. For the twelve sites at hex-distance two from a fixed site, each carrying one lamp of the same sign, the support of the flow consists of two vertex-disjoint cycles, so $\mathrm{st} \geq 1$ for every translate of the configuration, although the site set is hex-connected; translated to contain a base site it has $\mathrm{st} = 1$ and $\ell = 48 + 2 = 50$. Connectivity of the site set therefore does not suffice, and the support condition of Corollary 5.10 cannot be weakened.

The number of translations of each word length for flow-generic τ is, as computed below,

$$6,\ 6,\ 42,\ 96,\ 350,\ 1092,\ 3684$$

at lengths $6, 8, 10, 12, 14, 16, 18$ respectively, and odd lengths carry none, as the theorem requires: enumeration of the configurations with $\|\phi(c)\|_1 + 2\,\mathrm{st}(c) \leq 18$, over the bounded region described in `code/triangle_relations/honeycomb_metric_and_` yields these counts, and exact rational breadth-first search at the witness with apex $(2/7, 3/5)$ returns the same sets of translation vectors, element by element at every length, with distinct configurations receiving distinct vectors. At the witness with apex $(1/3, 1/2)$ the same search returns 3680 at length 18: that shape is not flow-generic, and four coincidences mask the count there. The configurations with one positive lamp at each of $(0,-2), (0,-1), (1,-2), (2,-2)$ and at each of $(0,1), (1,1), (2,0), (2,1)$, of length 18 at flow-generic shapes, represent the same translation at that witness, as do the configurations on $(-2,-1), (-2,0), (-1,-1), (0,-1)$ and on $(-2,2), (-1,2), (0,1), (0,2)$, together with the inverses of both pairs; at every other length the two witnesses agree. Of the 500 translations of length at most 14, the values $\mathrm{st} = 0, 1, 2, 3, 4$ occur $264, 152, 54, 12, 18$ times.

6 Non-generic strata

Proposition 6.1. *The growth series of* $W_m = \langle x_0, x_1, x_2 \mid x_i^2, (x_0x_1)^m \rangle$ *with respect to* x_0, x_1, x_2 *is*

$$W_m(t) = \frac{(1+t)\,(1+t+\cdots+t^{m-1})}{1-t-t^2-\cdots-t^m},$$

so its growth rate is the m*-bonacci constant.*

Proof. Steinberg's formula [1, 2] for a Coxeter system, which sums over the standard parabolics with finite W_J, applied here with one finite rank two parabolic of order $2m$ and two infinite bonds, gives $1/W_m(t^{-1}) = 1-3/(1+t)+1/\big((1+t)(1+t+\cdots+t^{m-1})\big)$; solving for W_m yields the displayed expression. □

Let one angle of τ equal π/m, the shape being otherwise generic, and let $d^*(m)$ be the first depth at which u_d falls below $[t^d]W_m(t)$, with deficit $\delta(m)$ there. Exact computation in $\mathbb{Q}(2\cos(\pi/m))$, at two rational choices of the legs for each m, gives

m	2	3	4	5	6	7	8	9	10	11	12
d^*	10	9	6	11	8	11	10	11	11	11	11
δ	8	3	3	33	8	33	15	33	33	33	33

The entries $(11, 33)$ are the universal deficit of Theorem 1.1. Beyond the universal values, the deficit at depth 12 exceeds the generic 132 by 6 for $m = 5$, by 8 for $m = 9$ and by 24 for $m = 10$, and equals it for $m = 7, 11, 12$. Every entry of the table and every one of those deficits is regenerated by `code/zeta_probe/tools/paper4_strata`, part (A), at up to six leg samples per m and at two primes.

The onset is therefore earlier than the universal depth 11 exactly for $m \in \{2, 3, 4, 6, 8\}$ in this range. The crystallographic orders are not distinguished: $m = 8$ has onset 10, and the mechanism is instead one of depth, as the following theorem shows.

Theorem 6.2 (Rotation relations)**.** *Let* $m \geq 3$ *and let* τ *have angle* π/m *at the vertex where the sides* 0 *and* 1 *meet, the shape being otherwise arbitrary. For a word* w *in* r_0, r_1, r_2 *and a letter* i*, let* $c_i(w)$ *be the number of occurrences of* i *at odd positions of* w *minus the number at even positions. Fix* $1 \leq c \leq m-1$ *and put* $n = m/\gcd(c, m)$*.*

(i) *Let w have even length. Its linear part is, at every shape of the stratum, the rotation by $2\pi c_1(w)/m$ if and only if $c_2(w) = 0$; such a w represents a rotation of order $m/\gcd(c_1(w), m)$ if $m \nmid c_1(w)$, and a translation otherwise.*

(ii) *The reduced words of length $2c+2$ with $c_2 = 0$ and $c_1 = c$ are exactly the c^2 words $w_{a,b}$ with $0 \le a \le c$, $1 \le b \le c+1$ and $b \notin \{a, a+1\}$, in which every odd position carries 1 except position $2a+1$, carrying 2, and every even position carries 0 except position $2b$, carrying 2. Reversal maps them bijectively onto the reduced words of length $2c+2$ with $c_2 = 0$ and $c_1 = -c$, and no shorter word of even length has $c_2 = 0$ and $|c_1| = c$ except $(r_0r_1)^c$ and its reversal.*

(iii) *Exactly one of the $w_{a,b}$, namely $w_{0,c+1} = r_2(r_0r_1)^c r_2$, has finite order in W_m. Each of the other c^2-1 satisfies $w^n = 1$ throughout the stratum without this being a consequence of the Coxeter relations, and w^n is cyclically reduced of length $n(2c+2)$, so splitting it in half exhibits a coincidence between two reduced words of length $n(c+1)$; they are distinct, since equality in F would give $w^n = 1$ there. Whether the coincidence lowers the sphere at that depth requires the two halves to be geodesic in W_m; this holds in the range $c \le m/2$, where every $\{0,1\}$-block of $w_{a,b}$, including the join u_2u_0, has length at most m, and can fail for larger c. Every case of Corollary 6.3 lies in that range except the prime branch at $m = 3$, where $c = 2 > m/2$; see Remark 6.4.*

Corollary 6.3. *Let d be a divisor of m with $2 \le d \le m/2$. The stratum carries $d^2 - 1$ relations at depth $m + m/d$. Taking d largest, $d^*(m) \le m + p$ with p the least prime factor of m, the family having $(m/p)^2 - 1$ members. For m prime no such d exists, and minimising the depth $(c+1)m/\gcd(c,m)$ of Theorem 6.2 over $2 \le c \le m-1$ instead gives $c = 2$, so $d^*(m) \le 3m$ with 3 members. For even m this is $d^*(m) \le m+2$ with $(m/2)^2 - 1$ members, and the relations are half-turns, $n = 2$ making each relation $w = \overline{w}$.*

Proof of Theorem 6.2. (i) For even L the linear part of $r_{a_1} \cdots r_{a_L}$ is the rotation by $2\sum_j (-1)^{j+1}\theta_{a_j} = 2\big(c_0\theta_0 + c_1\theta_1 + c_2\theta_2\big)$, with θ_i the direction of the ith side; normalize $\theta_0 = 0$ and $\theta_1 = \pi/m$. As the shape runs through the stratum θ_2 runs through an interval while θ_0, θ_1 stay fixed, so the linear part is the same at every shape of the stratum precisely when $c_2 = 0$, and then it is the rotation by $2\pi c_1/m$. A plane isometry whose linear part is a nontrivial rotation fixes a point and is the rotation by that angle about it, whence the stated order; if the linear part is trivial the isometry is a translation.

(ii) Put $L = 2c+2$ and let p, q count the letters 1 at odd and at even positions, so $p - q = c_1 = c$. Each parity class has $c+1$ positions, so $p \le c+1$ and hence $q \le 1$. If $q = 1$ then $p = c+1$, so every odd position carries 1; the even position carrying 1 has a neighbour, necessarily odd and carrying 1, contradicting reducedness. So $q = 0$ and $p = c$: no even position carries 1, and exactly one odd position carries another letter. Were that letter 0, then no letter 2 would occur at an odd position, hence none at an even position either since $c_2 = 0$, so every even position would carry 0 and the exceptional odd position would have a neighbour carrying 0, again contradicting reducedness. Hence it carries 2, and $c_2 = 0$ forces exactly one even position to carry 2, the others carrying 0. This is the displayed shape. Letters 2 at positions $2a+1$ and $2b$ are adjacent exactly when $b \in \{a, a+1\}$, and no other neighbouring letters coincide, so the admissible pairs are those listed, $(c+1)^2 - (2c+1) = c^2$ in number. Reversal of a word of even length exchanges the two parity classes and so negates every c_i. For the last clause, $|c_1| = c$ requires at least c letters 1 in one parity class, so an even length is at least $2c$; at length $2c$ the argument just given leaves only $(r_0 r_1)^c$ and its reversal. (For odd lengths the linear part is a reflection and (i) does not apply.)

(iii) By (i) each $w_{a,b}$ is a rotation of order n at every shape of the stratum, so $w_{a,b}^n = 1$ there. Since the Coxeter presentation of W_m imposes no relation involving x_2 beyond x_2^2,

$$W_m \;\cong\; D_m * C_2, \qquad D_m = \langle x_0, x_1 \rangle, \quad C_2 = \langle x_2 \rangle,$$

and by the torsion theorem for free products [3, Thm. 4.5] (see also [4, §I.1.3]) every element of finite order is conjugate into a factor; equivalently a cyclically reduced element with two or more syllables from D_m has infinite order. Write $w_{a,b} = u_0 x_2 u_1 x_2 u_2$ with $u_i \in D_m$ the images of its maximal $\{0,1\}$-blocks. Its cyclic reduction is $u_2 u_0\, x_2 u_1 x_2$, so, u_1 being nontrivial by reducedness, $w_{a,b}$ has finite order if and only if $u_2 = u_0^{-1}$. Suppose first $b \ge a+2$. Reading the blocks gives $u_0 = (x_1x_0)^a$, $u_1 = (x_0x_1)^{b-a-1}$ and $u_2 = (x_1x_0)^{c+1-b}$, so $u_2 = u_0^{-1}$ amounts to $b \equiv a+c+1 \pmod{m}$; as $2 \le b-a \le c+1 \le m$ this forces $b-a = c+1$, that is $a = 0$ and $b = c+1$, where indeed $w_{0,c+1} = r_2(r_0r_1)^c r_2$ is conjugate to $(r_0r_1)^c$. Suppose next $b \le a-1$. Then $u_0 = (x_1x_0)^{b-1}x_1$, $u_1 = (x_1x_0)^{a-b}$ and $u_2 = x_0(x_1x_0)^{c-a}$; writing $s = x_0x_1$ and $x_1 = x_0 s$, the condition $u_2 = u_0^{-1}$ becomes $x_0 s^{a-c} = x_0 s^b$, that is $b \equiv a - c \pmod{m}$, incompatible with $1 \le b \le a-1 \le c-1$. Finally the first letter of $w_{a,b}$ is 1 unless $a = 0$ and the last is 0 unless $b = c+1$, so for the remaining $c^2 - 1$ words no cancellation occurs in $w_{a,b}^n$, which is therefore cyclically reduced of length $n(2c+2)$. □

Proof of Corollary 6.3. Apply Theorem 6.2 with $c = d$. A divisor $d \geq 2$ of m with $d \neq m$ satisfies $m/d \geq 2$, hence $d \leq m/2 \leq m - 1$, so $c = d$ is in the range of the theorem; and $\gcd(d, m) = d$, so $n = m/d$. Part (iii) then gives $d^2 - 1$ words, each contributing a coincidence between two reduced words of length

$$n(c+1) \;=\; \frac{m}{d}\,(d+1) \;=\; m + \frac{m}{d}.$$

That depth decreases as d increases, so among the divisors it is least at the largest proper divisor $d = m/p$, with p the least prime factor of m: depth $m + p$ and $(m/p)^2 - 1$ relations. For even $m \geq 4$ this is $p = 2$, $d = m/2$, depth $m + 2$ and $(m/2)^2 - 1$ relations; there $n = 2$, so $w_{a,b}$ is a half-turn, $w_{a,b}^2 = 1$, and the coincidence obtained by splitting $w_{a,b}^2$ in half is $w_{a,b} = \overline{w_{a,b}}$, the reversal being the inverse word since the generators are involutions.

If m is prime it has no divisor in $[2, m/2]$, and $\gcd(c, m) = 1$ for every $1 \leq c \leq m - 1$, so $n = m$ and the depth of Theorem 6.2 is $m(c + 1)$ throughout. It is increasing in c, and $c = 1$ contributes $c^2 - 1 = 0$ relations, so the least depth carrying relations is at $c = 2$: depth $3m$, three relations. □

Remark 6.4. The prime branch uses $c = 2$, which lies in the range $c \leq m/2$ of Theorem 6.2(iii) for every prime $m \geq 5$ but not for $m = 3$, where $m/2 = 3/2$. At $m = 3$ the corollary therefore produces three coincidences at depth 9 without the theorem certifying that their halves are geodesic in W_3, so the entry $d^*(3) = 9$ of the table rests on the computation and not on the corollary. The branch is inside the range for $m = 5, 7, 11$, and so is every case of the divisor branch.

The counts $3, 8, 15, 24$ at depths $6, 8, 10, 12$ for $m = 4, 6, 8, 10$ agree with Corollary 6.3; $m = 2$ gives none, and $m = 12$ is placed at depth 14 with 35 relations, beyond the computed window and consistent with the absence of extra relations at depth 12. That placement should not be read as saying that depth 14 is where the $m = 12$ stratum first departs from the generic growth. Measured against the generic spheres 11925 and 23570, the $m = 12$ stratum has 11878 elements at depth 13 and 23358 at depth 14, deficits of 47 and 212 (part (A) of `paper4_strata` again); the deficit at depth 13 is of the second kind discussed below and is not accounted for by Corollary 6.3. For odd m the corollary gives 3 relations at depth 9 for $m = 3$ and 8 at depth 12 for $m = 9$, both attained, and for the primes $m = 5, 7, 11$ it gives 3 relations at depths $15, 21, 33$. The bound is attained for every computed onset except two. For $m = 2$ the theorem produces nothing, and for $m = 5$ six relations occur already at depth 12, before the predicted 15. Of those six, three equate two translations: on the stratum $(r_0 r_1)^m = 1$ identifies $t_{n,j}$ with $t_{n+m,j}$, so the site lattice of §3 wraps into the cylinder $(\mathbb{Z}/m)\times\mathbb{Z}$ and the flows of §5 live on a cylindrical honeycomb; the remaining three have $c_2 = \pm 1$. Theorem 6.6 below accounts for the translation kind: they are pairs of words whose flows differ by a wrapping cycle. We do not count them here, and for $m = 7$ and $m = 11$ we do not know whether relations of this second kind precede the depths 21 and 33 of the theorem. That the relations of Theorem 6.2 are the only coincidences at their depth, so that δ equals the stated count rather than merely dominating it, is the computational input recorded above.

For $m = 3, 4, 6, 8$ the relations tabulated above hold on the whole stratum: each is verified identically in the legs (a, b) over the relevant field, so the onset is a property of the stratum and not of the samples, while the deficit being exactly δ is witnessed at the rational samples and therefore holds off a proper closed subset of the stratum. The row $m = 2$ is verified at three rational samples. The three relations at depth 9 for $m = 3$ are

$$210201210 = 102012102, \quad 020121020 = 121020121, \quad 201210201 = 012102012,$$

of which the first and third have the form $R\,s\,R\,s^{-1}R\,s$, the third with s replaced throughout by s^{-1}, with $R = r_2$ and $s = r_1 r_0$ the apex rotation of order three, while the second has as relator the cube $(r_0r_2r_0r_1r_2r_1)^3$. The three at depth 6 for $m = 4$ are

$$012102 = 201210, \quad 020121 = 121020, \quad 102012 = 210201.$$

6.1 The translation subgroup on a stratum

Let τ be generic on the stratum $\alpha = \pi/m$ and put $\widehat{P}_m = \widehat{P}/\langle \rho_1^m \rangle \cong (\mathbb{Z}/m \times \mathbb{Z}) \rtimes C_2$, with Cayley graph X_m: the quotient of the honeycomb X by the m-fold translation, a cylinder. Its faces are the wrapped sites $(n \bmod m,\, j)$, and the constructions of §5 apply to X_m unchanged, a word representing a translation having a closed suffix walk whose flow is a finitely supported cycle on X_m with $t(w) = V_\tau(\phi(w))$.

Definition 6.5. Call τ *stratum-generic* for $\alpha = \pi/m$ if M_τ is injective on $\widehat{P}_m$. Since M_τ is injective on the rotation subgroup exactly when $a\,(2\pi/m) + b\,\beta_2 \notin 2\pi\mathbb{Z}$ for $(a \bmod m, b) \neq (0, 0)$, and each failure is a nontrivial condition on the remaining shape parameter, this holds off countably many proper closed subsets of the stratum.

Theorem 6.6 (Stratum metric)**.** *Let τ be stratum-generic for $\alpha = \pi/m$.*

(i) $\sum_{n\in\mathbb{Z}/m} t_{n,j} = 0$ *for every j: the m lamps of a level sum to zero, so the wrapped lamps are not independent.*

(ii) The flow of a word representing a given translation is no longer unique. The word $(r_0r_1)^m$ and its conjugates have closed suffix walks, whose flows wrap once around the cylinder and are killed by V_τ. Write γ_j for the flow of $((r_0r_1)^m)(r_0r_2)^j$, the conjugate by the jth power of r_0r_2. If moreover $2 \le m \le 12$, then $\gamma_j - \gamma_{j-1}$ is the sum of the m faces of level j.

(iii) For every translation g, writing $Z_1(X_m)$ for the finitely supported 1-cycles of X_m and extending st *from configurations to cycles by* $\mathrm{st}(\phi) := \mathrm{st}(c)$ *for any configuration c with flow ϕ,*

$$\ell(g) \;=\; \min\bigl\{\, \|\phi\|_1 + 2\,\mathrm{st}(\phi) \;:\; \phi \in Z_1(X_m),\; V_\tau(\phi) = g \,\bigr\}.$$

Proof. (i) Conjugation by r_0r_1 is the rotation by $2\pi/m$ about the apex, so it rotates the translation vector of a lamp by that angle; the m vectors $t_{n,j}$ with $n \in \mathbb{Z}/m$ are therefore the images of one vector under the mth roots of unity, and those sum to zero. (ii) $(r_0r_1)^m$ is the rotation by 2π, that is the identity, so the flow of its word has $V_\tau = 0$. For the identity between $\gamma_j - \gamma_{j-1}$ and the level sum, both sides are explicit finitely supported flows, and conjugation by r_0r_2 carries level j to level $j+1$ and γ_j to γ_{j+1}, so at a fixed m it is enough to compare them at a single j, which is a finite computation. It is carried out for $2 \le m \le 12$ in `code/triangle_relations/cylinder_structure.py` and is why the statement carries that range; uniformly in m it is a conjecture, and nothing below uses it, part (iii) being independent of it. (iii) Neither bound of Theorem 5.7 used the planarity of X or the uniqueness of the flow. The lower bound gives $\ell(w) \ge \|\phi(w)\|_1 + 2\,\mathrm{st}(\phi(w))$ for every word w representing g, and $\phi(w)$ is one of the competitors; conversely the Eulerian construction turns any competitor ϕ into a word of length $\|\phi\|_1 + 2\,\mathrm{st}(\phi)$ with flow ϕ, representing $V_\tau(\phi) = g$. □

For $m = 5$ the formula was checked against exact rational breadth-first search on all 158 translations of length at most 12, the minimum being attained already among the flows differing from $\phi(w)$ by a single γ_j. That stratum has $6, 6, 42, 104$ translations at lengths $6, 8, 10, 12$, against $6, 6, 42, 96$ for a generic shape: wrapping shortens elements as well as identifying them.

The identifications themselves can be counted. Sending each of the 5276 generic translations of length at most 18 to its image on the stratum, the first coincidence appears at generic length $2m + 4$ for $m = 3, 5, 7$, that is at $10, 14, 18$. For $m = 5$ and $m = 7$ it is the plain wrap, the sites $(2, 0)$ and $(-3, 0)$ for $m = 5$, and $(3, 0)$ and $(-4, 0)$ for $m = 7$, which are identified by the cylinder and are the first such pair of equal generic length; for $m = 3$ it is instead the level relation of part (i), which equates the lamp $t_{0,0}$ of length 6 with the configuration $-t_{-2,0} - t_{-1,0}$ of length 10. Counting them requires two conventions to be fixed, because neither is implied by the word "identification". First, we count the *excess*, the number $5276 - \#\{\text{distinct stratum images}\}$, rather than the number of unordered colliding pairs; the two differ by more than a factor of four at $m = 3$. Second, and less obviously, the count is *not* constant along the stratum, so a leg sample must be named. At the leg ratio $2 : 3$ the excesses are 3255 for $m = 3$, 476 for $m = 5$, 14 for $m = 7$, and none for $m = 9$ and $m = 11$; for $m = 3$ moreover 76 generic translations become trivial. At the ratio $1 : 3$ the same computation gives 3241 and 302 for $m = 3, 5$, and at the ratio $1 : 2$ it gives 64 for $m = 9$ in place of 0; the ratio $1 : 2$ at $m = 3$ is the $\pi/6, \pi/3, \pi/2$ triangle and degenerates completely, leaving 37 classes. These counts are therefore properties of a shape, not of the stratum, and the hazard of Remark 6.15 applies to them as much as to the census. All of them, and the 5276, are the output of `code/zeta_probe/tools/strat_ident`.

These identifications are of two kinds. Those between configurations with the same wrapped configuration hold already in W_m, where $(x_0x_1)^m = 1$ forces $t_{n,j} = t_{n+m,j}$, and are not new relations; the rest come from the level relation of Theorem 6.6(i) and are new. We do not report a numerical split between the two kinds. Deciding it requires the lamp configuration of each translation, not merely its value: W_m is a quotient of the free product but not of the generic group, so a generic translation has no well-defined image in W_m against which to test. An earlier version of this paper reported a split whose two parts did not sum to the total. The first identification of the second kind is in each case one level split into two arcs,

$$t_{0,0} = -t_{1,0} - t_{2,0}\ (m = 3), \quad t_{3,0} + t_{4,0} = -\sum_{n\le 2} t_{n,0}\ (m = 5), \quad \sum_{n\le 2} t_{n,0} = -\sum_{n\ge 3} t_{n,0}\ (m = 7),$$

at generic lengths 6 and 10, 10 and 14, 14 and 18 respectively. An arc of k consecutive lamps of one sign has length $4k + 2$, by Corollary 5.9 and Corollary 5.10, so the balanced split of a level of m lamps first becomes visible at depth $4\lceil m/2\rceil + 2$, which is $2m + 4$ for odd m and $2m + 2$ for even m.

Comparing with Corollary 6.3: for even m the rotation relations at depth $m+2$ come first, and for odd composite m so does the divisor bound $m+p$; for $m = 3$ the rotation depth 9 precedes 10. But for every prime $m \ge 5$ one has $2m+4 < 3m$, so there the level relations arrive first, at depths $14, 18, 26$ for $m = 5, 7, 11$. The stratum also carries translations that are not images of generic translations, a word being able to have trivial linear part on the stratum without having one generically. Both effects are accounted for by the following census.

Lemma 6.7 (Closed form for k)**.** *For every site (n, j),*

$$k(n,j) = \begin{cases} 2n+2j-1, & j \geq 1,\ n \geq 0, \\ 2j-2, & j \geq 1,\ -j \leq n \leq -1, \\ -2n-3, & j \geq 1,\ n \leq -j-1, \\ 2n, & j \leq 0,\ n \geq -j, \\ -2j-1, & j \leq 0,\ 0 \leq n \leq -j-1, \\ -2n-2j-2, & j \leq 0,\ n \leq -1. \end{cases}$$

Each row therefore has a single flat valley, of width j at $[-j, -1]$ with value $2j-2$ when $j \geq 1$ and of width $-j$ at $[0, -j-1]$ with value $-2j-1$ when $j \leq -1$, the value rising by 2 per step outside it; the row minimum is 0 exactly for $j \in \{0, 1\}$. The rows $j = 0$ and $j = 1$ recover $k(n, 0) = 2n$, $-2n-2$ and $k(n, 1) = 2n+1$, $k(-1, 1) = 0$, $-2n-3$.

Proof of Lemma 6.7. We first give the distance on the *vertices* of X and then minimise over the six corners of a hexagon. Recall that a vertex of X is a triangle of mutually hex-adjacent sites, of one of the two families

$$\mathrm{up}(n,j) = \{(n,j), (n{+}1,j), (n,j{+}1)\}, \qquad \mathrm{down}(n,j) = \{(n,j), (n{+}1,j), (n{+}1,j{-}1)\},$$

that $\mathrm{up}(n, j)$ is adjacent exactly to $\mathrm{down}(n, j)$, $\mathrm{down}(n-1, j+1)$ and $\mathrm{down}(n, j+1)$, and that the base vertex is $e = \mathrm{up}(-1, 0)$, the triangle of the three base sites. Put

$$\widehat{\delta}_0(n,j) = \max\bigl(-2n-2+2j^-,\ 2|j|,\ 2n+2+2j^+\bigr), \qquad j^\pm = \max(0, \pm j),$$

and $\widehat{\delta}(\mathrm{up}(n, j)) = \widehat{\delta}_0(n, j)$, $\widehat{\delta}(\mathrm{down}(n, j)) = \widehat{\delta}_0(n, j)-1$ if $j \geq 1$ and $\widehat{\delta}_0(n, j)+1$ if $j \leq 0$.

We use the local criterion for a distance function: if $\widehat{\delta}(e) = 0$, if $|\widehat{\delta}(v) - \widehat{\delta}(w)| \leq 1$ across every edge, and if every vertex other than e has a neighbour at which $\widehat{\delta}$ is smaller by one, then $\widehat{\delta} = d_X(e, \cdot)$. The second condition gives $d_X \geq \widehat{\delta}$ along any geodesic and the third gives $d_X \leq \widehat{\delta}$ by induction on $\widehat{\delta}$.

The first condition is immediate. The other two are *finitely many* linear inequalities, and this is the point of writing $\widehat{\delta}_0$ as a maximum of three affine functions: it is piecewise affine with regions cut out by the lines $j = 0$, $n = -1$ and $n = -j-1$, the branch on the family adds only the sign of j, and each neighbour shifts (n, j) by at most one in each coordinate. Hence whether a given local inequality holds depends only on which pair of regions the two endpoints lie in, of which there are finitely many, and every such pair is already realised among the sites with $|n|, |j| \leq 44$. Checking the conditions there therefore checks every case. That check is carried out in `code/zeta_probe/tools/hexdist`: across all 47 526 edges of the window the largest change in $\widehat{\delta}$ is 1, and no vertex other than e lacks a descending neighbour. So $\widehat{\delta} = d_X(e, \cdot)$.

Finally $k(n, j)$ is the minimum of $\widehat{\delta}$ over the six corners of the hexagon at (n, j), namely $\mathrm{up}(n, j)$, $\mathrm{up}(n-1, j)$, $\mathrm{up}(n, j-1)$, $\mathrm{down}(n, j)$, $\mathrm{down}(n-1, j)$ and $\mathrm{down}(n-1, j+1)$. Evaluating that minimum region by region gives the displayed closed form. □

Remark 6.8 (what the proof rests on). The lemma is machine-checked as a statement about the graph distance d_X, in three files under `lean/with_mathlib/`.

`GraphLocalDistance.lean` proves the local criterion in the generality in which it is true. For an arbitrary `SimpleGraph` G, a vertex e and a function f from the vertices to the non-negative integers: if $f(e) = 0$, if $f(w) \leq f(v)+1$ across every edge $\{v, w\}$, and if every vertex $v \neq e$ has a neighbour w with $f(w) + 1 = f(v)$, then $f = d_G(e, \cdot)$. This is `dist_eq_of_base_lip_descent`; no finiteness, local finiteness or connectedness is assumed, the descent hypothesis itself supplying reachability.

`HexGraph.lean` builds X as a `SimpleGraph` on $\mathbb{Z} \times \mathbb{Z} \times \{\mathrm{up}, \mathrm{down}\}$ and discharges the three hypotheses. The adjacency is not taken on trust: `adj_iff_sharesEdge` proves that two triangles are adjacent exactly when they are distinct and have two sites in common, so X is the share-an-edge graph on the triangles of the site lattice, and `memTri_iff` proves that the triangles containing a given site are exactly the six corners listed above. The conclusions are `dist_up` and `dist_down`, which give $d_X(e, \cdot) = \widehat{\delta}$ at every vertex, and `lem_krows_face`, which is the displayed closed form for $\inf\{d_X(e, v) : v \text{ a vertex of } f_{n,j}\}$, that is for $k(n, j)$ as defined in §4. Connectedness of X falls out as `X_connected`.

`HexDistance.lean` carries the arithmetic to which the criterion reduces the geometry: the base value `dhat_base`, the three Lipschitz bounds `lip_same`, `lip_left`, `lip_up`, the two descent statements `descent_down`, `descent_up`, and the evaluation of the corner minimum `lem_krows`. These are proved for all integers n, j by case analysis on the affine regions, so the reduction to a finite window in the proof above is a convenience of exposition and not a step the verification depends on.

All three files build with no `sorry` and with every `#print axioms` reporting only `propext`, `Classical.choice` and `Quot.sound`. One step is not machine-checked, and cannot be: the reading of the definition of X in §5 into the Lean construction, namely that the honeycomb there is the graph on triangles of mutually hex-adjacent sites with base vertex the triangle $\{(0,0),(-1,0),(-1,1)\}$ of base sites, and that the face $f_{n,j}$ is its set of triangles containing (n,j).

The exhaustive search of `code/zeta_probe/tools/hexdist` remains as an independent check of the same statement. It builds X from the site lattice directly and compares breadth-first search against the closed form at every site with $|n|,|j| \leq 60$, that is 14 641 sites, with no exception; on a window of 16 562 vertices and 47 526 edges the largest change in $\widehat{\delta}$ across an edge is 1 and no vertex other than e lacks a descending neighbour. It also records the structural fact behind the form: writing $D(n,j)$ for the hex-lattice distance from (n,j) to the base-site set, one has $k = 2D - \varepsilon$ with $\varepsilon \in \{0,1\}$, and $\varepsilon = 0$ exactly on the valley when $j \geq 1$ and exactly off it when $j \leq -1$. The lemma is stated as one identity rather than as two base rows because everything below uses only the identity, and because a single verified statement is easier to audit than a proof given on two rows and extrapolated silently to the others.

The three involutions σ_i each reverse the sign of n and shift it, so a non-backtracking walk alternates the two orientations and moves $|n|$ by one per two steps, changing j only through σ_0, σ_1; that accounts for the slope 2 away from the valley, and the rows $j = 0, 1$ have minimum 0 because they contain a base site while no other row does. It does not by itself pin the valley widths, which is why Remark 6.8 states the closed form as verified.

Remark 6.9. The rows off $j \in \{0,1\}$ are *not* V-shaped with slope 2; this is what defeats the argument used in an earlier version of Corollary 6.10. Their minima are attained on flat valleys whose width grows linearly in $|j|$: on $j = 3$ the values for $n = -4, \dots, 4$ are $5, 4, 4, 4, 5, 7, 9, 11, 13$, so the minimum 4 is attained three times and the first step out of the valley is $+1$, not $+2$. The case analysis in the proof below replaces that argument and does not assume any shape beyond the closed form itself.

Corollary 6.10. *Assume Lemma 6.7. Then for every $h \geq 2$,*

$$\min_{(n,j)} \max\{k(n,j), k(n+h,j)\} = h - 1,$$

attained on the row $j = 1$ at $n = -h/2 - 1$ when h is even and on the row $j = 0$ at $n = -(h+1)/2$ when h is odd.

Proof. The lower bound. Assume Lemma 6.7, let $h \geq 2$, and write $K := \max\{k(n,j), k(n+h,j)\}$.

Let $j \leq 0$ and put $M := -j \geq 0$, so k equals $2n$ for $n \geq M$, equals $2M-1$ on $[0, M-1]$ and equals $-2n+2M-2$ for $n \leq -1$. If both points lie in $[0, M-1]$ then $h \leq M-1$ and $K = 2M-1 \geq 2h+1$. If $n \geq 0$ and $n+h \geq M$ then $K \geq 2(n+h) \geq 2h$. If $n+h \leq -1$ then $n \leq -h-1$ and $K \geq -2n+2M-2 \geq 2h+2M$. If $n \leq -1$ and $n+h \leq M-1$ then $-n \geq h-M+1$, so $K \geq -2n+2M-2 \geq 2h$. If $n \leq -1$ and $n+h \geq M$ the two values sum to $(-2n+2M-2)+(2n+2h) = 2M+2h-2$, so $K \geq M+h-1$. If $0 \leq n \leq M-1$ and $n+h \geq M$ then $K \geq 2(n+h) \geq 2h$. Every case gives $K \geq h-1$.

Let $j \geq 1$, so k equals $2n+2j-1$ for $n \geq 0$, equals $2j-2$ on $[-j, -1]$ and equals $-2n-3$ for $n \leq -j-1$. If both points lie in $[-j, -1]$ then $h \leq j-1$ and $K = 2j-2 \geq 2h$. If $n \geq 0$ then $K \geq 2(n+h)+2j-1 \geq 2h+1$. If $n+h \leq -j-1$ then $n \leq -j-1-h$ and $K \geq -2n-3 \geq 2j+2h-1$. If $n \leq -j-1$ and $-j \leq n+h \leq -1$ then $-n \geq h+1$, so $K \geq -2n-3 \geq 2h-1$. If $n \leq -j-1$ and $n+h \geq 0$ the two values sum to $2h+2j-4$, so $K \geq h+j-2$. If $-j \leq n \leq -1$ and $n+h \geq 0$ then $n+h \geq h-j$, so $K \geq 2(h-j)+2j-1 = 2h-1$. Again every case gives $K \geq h-1$.

Attainment is the following computation with the row formulas of Lemma 6.7. On row 0 the pair $(n, n+h)$ with $n \leq -1 \leq 0 \leq n+h$ has values $-2n-2$ and $2(n+h)$, equal to $h-1$ exactly when h is odd and $n = -(h+1)/2$; on row 1 the values are $-2n-3$ and $2(n+h)+1$, equal to $h-1$ exactly when h is even and $n = -h/2-1$. Parity makes one of the two rows attain $h-1$ for every h. Both halves are machine-checked in `lean/with_mathlib/AntipairRows.lean`, and against the closed form itself rather than against a restatement of it: the file imports `HexDistance.kClosed`, the function of Lemma 6.7, and proves attainment as `attains_kClosed`, the lower bound as `antipair_lower_bound_kClosed`, and the corollary as the single statement `antipair_min_kClosed`, an `IsLeast` asserting that $h-1$ is attained and is a lower bound. The file also refutes the premise of the withdrawn proof, again on `kClosed` (`row3_eq_kClosed`, `row3_valley_width_three_kClosed`, `row3_counterexample_kClosed`): by Remark 6.9 the rows off $j \in \{0, 1\}$ are not V-shaped with slope 2, which is why the valley argument fails and the case analysis above is needed in its place. $\square$

Remark 6.11. At $h = 1$ the statement still holds, with the value $h-1 = 0$: the lamp distance is nonnegative, and the sites $(-1, 0)$ and $(0, 0)$ both have $k = 0$. The case is separated out because the argument above uses $h \geq 2$, and it is what Proposition 6.12 needs at $m = 2$. It is machine-checked as `kClosed_nonneg` and `antipair_min_kClosed_one` in the same file.

Proposition 6.12 (Antipodal lamps, m even)**.** *Let m be even. Then $(r_0 r_1)^{m/2}$ is the half-turn about the apex, and*

$$t_{n+m/2,\,j} \;=\; -\,t_{n,j} \qquad \textit{for all } (n,j).$$

Part (i) of Theorem 6.6 follows by pairing n with $n + m/2$. The generic translations $t_{n,j}$ and $-t_{n+m/2,j}$, distinct for generic shapes, are identified on the stratum, and the shortest such identification has length $m+4$ for every even m. The earlier restriction to $m \le 20$ was forced by the lower bound of Corollary 6.10 being verified rather than proved; that bound is now proved from Lemma 6.7, so the statement holds for all even m on the same footing as the lemma, namely modulo the closed form.

Proof. $r_0 r_1$ is the rotation by $2\pi/m$ about the apex, so its $(m/2)$th power is the rotation by π. If h is a half-turn and t the translation by v, then hth^{-1} is the translation by $-v$. As $t_{n+m/2,j}$ is the conjugate of $t_{n,j}$ by $(r_0 r_1)^{m/2}$, the relation follows; summing it over the $m/2$ pairs $\{n, n+m/2\}$ of a level gives Theorem 6.6(i). A single lamp has length $6+2k(n,j)$ by Corollary 5.8, so an identification of this form first appears at $6+2\min\max\{k(n,j), k(n+m/2,j)\}$, the minimum over pairs. For $m \ge 4$ that minimum is $m/2-1$ by Corollary 6.10 with $h = m/2 \ge 2$, hence the length $m+4$. For $m=2$ the separation is $h=1$, outside the range of the corollary; there the minimum is 0 by Remark 6.11, and $6 = m+4$ again. ☐

For even $m \ge 4$ this precedes the length $2m+2$ at which new translations appear, so the identifications are what the census sees first, and it falls below the generic one; for odd m there is no such pairing and the census rises instead.

Proposition 6.13 (Stratum translation census)**.** *For τ generic on the stratum $\alpha = \pi/m$, the numbers of translations of each length are*

length	6	8	10	12	14	16	18
generic	6	6	42	96	350	1092	3684
$m=3$	6	8	38	76	224	630	1456
$m=4$	6	4	34	62	198	480	1394
$m=5$	6	6	42	104	448	1274	4612
$m=6$	6	6	40	78	278	750	2342
$m=7$	6	6	42	96	350	1110	4204
$m=9,11$	6	6	42	96	350	1092	3684

For odd $m \ge 3$ the two censuses agree up to length $2m+1$ and the stratum acquires translations that are not images of generic ones at $2m+2$; for even $m \ge 4$ they first differ at $m+4$, by Proposition 6.12.

Proof. By Theorem 6.2(i) a word represents a translation on the stratum exactly when $c_2 = 0$ and $m \mid c_1$. If $c_1 = 0$ the linear part is trivial at every shape, so the word represents a generic translation as well. Otherwise $|c_1| \geq m$, so by Theorem 6.2(ii) the word has length at least $2m$, with equality only for $(r_0r_1)^{\pm m}$, which is the identity on the stratum. Hence the shortest translation of the stratum that is not the image of a generic one has length $2m+2$. For odd m, identifications among generic translations require the level relation, hence length at least $4\lceil m/2\rceil+2 = 2m+4$ by the arc count above, while for even m Proposition 6.12 places the first identification at $m+4$. Shortening is excluded below $2m+2$ as well: if $g \neq 1$ has stratum length $L \leq 2m$, then a stratum-geodesic word for g has $c_1 = 0$, since otherwise $|c_1| \geq m$ forces length at least $2m$ with equality only at the identity; that word is then a word for a generic translation, so its generic length is at most L, and the reverse inequality is trivial. For odd m none of the three effects is available below $2m+2$ and the censuses agree length by length, while at $2m+2$ the first new translations appear. The displayed values are computed by breadth-first search in the stratum group, twice: in the number field $\mathbb{Q}(2\cos(\pi/m))(\sin(\pi/m))$ by `code/triangle_relations/rust_strat`, and in a finite field by `code/zeta_probe/tools/paper4_strata`, part (B). Every row is the profile taken at the majority of leg samples in the scan of Remark 6.14. The row $m = 9, 11$ reproduces the generic census exactly, as the onset statement requires, $2m+2$ exceeding 18 for both. □

Remark 6.14 (What the displayed table is). Two statements have to be separated here. That the census is a property of the stratum, constant off a finite set of leg parameters, is proved; the argument is given at the end of this remark. That the displayed values are that constant requires in addition that the samples used lie outside the exceptional set, and that is not proved but computed. The onset statements above are proved outright.

It is worth recording first what does *not* work, since an earlier version of this proposition offered it. A specialisation τ' of a stratum shape τ gives a surjection $G_\tau \twoheadrightarrow G_{\tau'}$, so *ball* counts at a sample bound the generic ones from below; but the table is of *sphere* counts, and those are not monotone, since an element of generic length 18 may have length 16 at τ' and so move between rows. Nor is the cumulative count bounded either way, because a word can acquire trivial linear part at τ' and enlarge $T_{\tau'}$ beyond the image of T_τ. Agreement at two samples is therefore evidence and not proof, and the hazard is real. Over the 43 coprime leg ratios $a{:}b$ with $a, b \leq 8$, the $m = 3$ stratum shows seven distinct profiles at lengths $6, \ldots, 18$ and the $m = 5$ stratum four; the tabulated row is in each case the one taken at most samples, 32 of 43 for $m = 3$ and 38 of 43 for $m = 5$, while for instance the legs 1:3 give $(224, 616, 1404)$ and 2:3 give $(224, 598, 1408)$ at lengths $14, 16, 18$ for $m = 3$, against the tabulated $(224, 630, 1456)$. The scan is `code/zeta_probe/tools/paper4_strata` in the mode `scan` m `18 8`. The argument that does close the first statement is the one used for Theorem 1.1, and we now give it.

Fix the length bound $L = 18$. There are finitely many reduced words of length at most L, hence finitely many ordered pairs of them. A stratum of fixed apex order m is one-parameter: after normalising, the shape is determined by the leg ratio t, and every matrix entry of $\rho_\tau(w)$ is a rational function of t with nonvanishing denominator. Each pair (w, w') therefore contributes a coincidence condition $\rho_\tau(w) = \rho_\tau(w')$, which after clearing denominators is the vanishing of finitely many polynomials in t; let $P_{w,w'}$ be their product. Two cases. If $P_{w,w'}$ vanishes identically then the coincidence holds at every shape of the stratum and belongs to the generic picture. If not, it vanishes at finitely many t. Let B be the union, over the finitely many pairs with $P_{w,w'} \not\equiv 0$, of those root sets; B is finite.

For $t \notin B$ the set of coincidences among words of length at most L is exactly the set of identically-vanishing conditions, independently of t. Every quantity read off from that set is therefore the same at every $t \notin B$: in particular the number of translations of each length at most L, which is what the table records. The displayed values are that common value, and they are computed at a witness. This also disposes of the two objections above, since no monotonicity in the specialisation is used: the argument compares generic shapes with each other, not a special shape with a generic one.

What the argument does not do is certify that a particular rational leg sample lies outside B, which is the second of the two statements separated at the start of this remark. That remains a computation, and it is why the samples are reported separately: their agreement is consistent with all of them being generic, and is not by itself a proof that any one is. The finiteness principle underlying all of this is machine-checked in `lean/with_mathlib/StratumGeneric.lean` (`badSet_finite`, `eval_eq_zero_iff_of_not_mem`, `count_eq_of_not_mem`, `exists_generic`), with only the standard axioms.

For six of the seven rows the sample can be dispensed with altogether. Fix the apex at 0 with the sides 0 and 1 through it at angle π/m, so that the first two reflections are $z \mapsto \bar{z}$ and $z \mapsto \omega^2\bar{z}$ with $\omega = e^{i\pi/m}$, and let the third side pass through 1 with direction an indeterminate V on the unit circle, $\bar{V} = V^{-1}$. Every element of the stratum group is then $z \mapsto aV^k z + b$ or $z \mapsto aV^k\bar{z} + b$ with coefficients in the Laurent ring $\mathbb{Z}[\omega][V, V^{-1}]$, and two words are equal in that ring exactly when they are equal at every shape of the stratum, while a difference that is a nonzero Laurent polynomial vanishes at only finitely many V. A breadth-first search in the ring therefore returns the generic sphere and the generic translation counts directly. That search is carried out in `lean/with_mathlib/CylCensus.lean`, and `census_m3`, `census_m4`, `census_m5`, `census_m6`, `census_m7` and `census_m9` reproduce the rows $m = 3, 4, 5, 6, 7, 9$ of the table exactly. Those six rows are therefore machine-checked as the generic values, at the cost of `native_decide` in the trusted base: each carries its own reflection axiom, so the Lean compiler is trusted alongside the kernel. The row $m = 11$ is not covered and remains computational.

For $m = 5$ the stratum has *more* translations than a generic shape at the length where they first differ, the new translations outnumbering the identifications; for $m = 3$, where the levels are short, the identifications dominate and the census falls below the generic one already at length 10. Beyond length 14 the tabulated values do not come from the configuration enumeration on X_m, which is not known to be exhaustive there and which for $m = 9$ undercounts; they come from the breadth-first search in the stratum group, and for $m = 9$ the correct value at length 18 is the generic 3684 by the argument of the proposition.

Remark 6.15. Rational leg samples that look generic need not be. For $m = 3$ the legs $(1, 2)$ give the right triangle with angles $\pi/6, \pi/3, \pi/2$, which collapses the census to $6, 0, 6, 6, 0, 12, 6$. Less visibly, for $m = 5$ the legs $(2, 3)$ carry 22 identifications at length 14 that the majority of samples do not: they give 426 translations there against the stratum-generic 448. Every computation reported here was repeated at three or more samples, and the leg scan of Remark 6.14 is what locates the exceptional ones.

Remark 6.16. These lines lie outside the *typical* stratum of [7], where the angles are $\mathbb{Q}$-linearly independent and the shortest non-generic relations have length 18. On the lines $\alpha = \pi/m$ the minimum drops as low as 12, attained by the relators of the three $m = 4$ relations displayed above.

Declarations

Funding. No funding was received for this work.

Competing interests. The author declares no competing interests.

Formalization. The arithmetic half of the lower bound of Theorem 5.7 is formalized in Lean 4 in `lean/TriangleFlowMetric.lean` of the cited repository, with no imports, no `sorry`, and no axioms beyond `propext` and `Quot.sound`. The correspondence is `flowNorm_le_traversals` and `traversals_parity` for the two per-edge facts, `connector_two_le` for the doubling of an edge carrying no flow, `steps_ge` for the summation, and `lower_bound` for the bound itself, the graph-theoretic half entering as its hypothesis `hst`. For the upper bound, `balanced` is the balance of the multigraph at each vertex, which is the cycle condition and is what makes the multigraph Eulerian, while `realize_flow`, `realize_traversals` and `upper_bound` give the flow and the length of the word read off a circuit. Euler's theorem itself, that a balanced multigraph whose edges are reachable from a base vertex admits a circuit using each directed edge once, is proved in `lean/EulerCircuit.lean` as `euler_circuit`, again with no imports and no `sorry`; Mathlib carries the necessary degree condition for Eulerian trails but not this sufficiency. The combinatorial content of both bounds of Theorem 5.7 is therefore machine-checked, while the identification of the translation subgroup with the flows is not formalized.

Theorem 6.2(ii) is formalized in `lean/RotationRelations.lean`, again with no imports and no `sorry`: `classify_normal_forms` is the classification of the reduced words of length $2c+2$ with $c_2 = 0$ and $c_1 = c$, `markBoth_valid` its converse, and `admissible_count` the count c^2. The free-product step behind part (iii) is formalized in `lean/with_mathlib/CoxeterTorsion.lean`, which does import Mathlib: writing a word of that shape as $u_0x_2u_1x_2u_2$ with the u_i in the dihedral factor, `finite_order_iff` states and proves that it has finite order in $W_m = D_m * C_2$ if and only if $u_2u_0 = 1$. Mathlib supplies the free product and its normal form but no torsion theorem for free products, so both directions are proved there, one by conjugating into the finite factor and the other by exhibiting every power as a nonempty reduced word. The step from that criterion to the identification of the unique word $w_{0,c+1}$ is formalized in `lean/with_mathlib/Bridge.lean` as `unique_finite_order`, for $c + 1 \leq m$, which is the range $c \leq m - 1$ used here; it is independently checked for $3 \leq m \leq 60$ and $1 \leq c \leq m - 1$ by `code/zeta_probe/tools/rust_torsion`.

Three further statements of Section 6 carry Lean certificates: the closed form of Lemma 6.7 in `lean/with_mathlib/HexDistance.lean`, Corollary 6.10 in `lean/with_mathlib/AntipairRows.lean`, and the finiteness principle behind Remark 6.14 in `lean/with_mathlib/StratumGeneric.lean`; the declaration names are given at the statements themselves. Everything named so far reports only `propext`, `Classical.choice` and `Quot.sound`, or a subset of the three.

A second tier exists and is kept separate because its trusted base is larger. Three files in lean/with_mathlib/ are proved by native_decide, so they trust the Lean compiler in addition to the kernel, each declaration carrying its own reflection axiom <thm>._native.native_decide.ax_1_1: CensusUniversal.lean (universal_identities: each of the 33 pairs of Theorem 1.1(i) as an identity in $\mathbb{Z}[X,Y]$), CensusWitness.lean (census_witness: the growth of a rational witness to radius 12, hence the deficits 33 and 132) and CylCensus.lean (census_m3 to census_m9: six rows of the table of Proposition 6.13, generically on the stratum, about twenty-one minutes of checking time and by far the slowest target in the build). Theorem 1.1 and Proposition 6.13 are asserted on the computations of the Reproducibility paragraph below; these three files are independent checks of them at the larger trusted base, the one place where the text leans on the second tier being the last paragraph of Remark 6.14, which says so.

Nothing else is formalized. In particular there is no certificate for the genericity half of Theorem 1.1(ii), for any of Section 3, for the identification of the translation subgroup with the flows in Section 5, or, in Section 6, for Proposition 6.1, Theorem 6.2(i), Corollary 6.3, Theorem 6.6 and the proof of Proposition 6.13. Absence of a named declaration means absence of a certificate.

Data availability. All code and verification data supporting the computational statements are openly available in the repository cited on the title page and archived on Zenodo (concept DOI 10.5281/zenodo.20370090).

Reproducibility. Every computed number in this paper is regenerated by a committed script; the index is code/triangle_relations/README.md, whose table maps each statement to the script that produces it and whose second section fixes the order in which the three chained scripts must be run. That directory holds the exact number fields $\mathbb{Q}(2\cos(\pi/m))(\sin(\pi/m))$ used on the strata, the flow and configuration model of §5, the whole-stratum certificates for $m = 4, 6, 8$, the verification of the 33 and the 132 coincidences as identities in $\mathbb{Z}[p,q]$, and the breadth-first searches. Five Rust tools sit under code/zeta_probe/tools/: hexdist (the local criterion and the closed form of Lemma 6.7), rust_torsion (the unique word of finite order, $3 \le m \le 60$), strat_ident (the identification counts of §6), paper4_ball12 (Theorem 1.1 and Remark 2.1 at depth 12) and paper4_strata (the table of d^* and δ, the sphere counts through depth 14, the stratum translation census, and the leg scan of Remark 6.14).

The last three work in a finite field $\mathbf{F}_p$, with $p \equiv 1$ modulo 55440 for paper4_strata, so that $\mathbf{F}_p$ contains a square root of -1 and a primitive $2m$th root of unity for every $m \le 12$ and the reflections are $\mathbf{F}_p$-rational.

Reduction modulo p can identify elements that are distinct in characteristic zero but never separate elements that are equal, so every count of distinct images they print is a lower bound for the count over the number field, and every deficit an upper bound. For the depth-12 census that turns the computation into a proof of the count: the matching upper bound is supplied by `verify_universal_relations.py`, which checks the 33 and the 132 coincidences as identities of polynomials in $\mathbb{Z}[p, q]$, so the 3039, the 6078 and the 6012 are exact. On the strata no such matching bound is available and the counts remain computational. The two `paper4_*` tools are run at two independent primes and agree exactly.

Use of AI tools. Large language models (Anthropic Claude) were used in the preparation of this manuscript, both in drafting and in computation. Every computational claim is certified by the scripts in the cited repository, independently of any model output. The author takes full responsibility for the content.

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